

Pratik J. Gandhi's CV

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Postdoctoral Associate, Department of Astronomy, Yale University

RESEARCH INTERESTS

theoretical astrophysics and galaxy formation
cosmological zoom-in simulations of galaxies
near-field cosmology and connections to cosmic reionisation
low-mass galaxies across cosmic time
stellar feedback and its impact on galaxy formation and evolution
galactic archeology and stellar fossil record of Local Group galaxies
synthetic observations and catalogs from galaxy simulations
pop III star formation, feedback, and enrichment

APPOINTMENTS

Postdoctoral Associate – Department of Astronomy, Yale University	Nov '24 – Present
Graduate Student – Department of Physics & Astronomy, UC Davis	Sep '18 – Oct '24

EDUCATION

Ph.D. in Physics (Astrophysics) – University of California, Davis	Sep '18 – Oct '24
M.S. in Physics (Astrophysics) – University of California, Davis	Sep '18 – June '20
B.S. in Astrophysics – Yale University	Aug '14 - May '18

FELLOWSHIPS & AWARDS

Outstanding Graduate Student Teaching Award UC Davis	May'23
FAMOUS travel grant AAS	Dec'22
Frontera Computational Science Fellowship TACC	Jun'22 - May'23
APS WiP Grant for UCD Diversity & Inclusion in Phys/Astro	Jan'21 - Jan'23
Graduate Program Research Fellowship UC Davis	Jan'21 - Apr'21
Paul Brady & Charles Soderquist Fellowship UC Davis	Sep'18 - Aug'19
Pierson College Mellon Forum Award Yale U.	Aug'17 - May'18
Tetelman Fellowship for International Research Yale U.	Summer 2016
Pierson College Richter Fellowship Yale U.	Summer 2016
Freshman Summer Research Fellowship Yale U.	Summer 2015

PUBLICATIONS, FIRST AUTHOR OR LEADING CONTRIBUTION

([ADS Library](#))

[5] **Gandhi, P. J.**, Bellinger, E., Geha, M., Wetzel, A., Hopkins, P. F., et al., “*Modelling stellar feedback from the first stars in the FIRE-3 cosmological zoom-in simulations*”, (in prep for submission to ApJ)

- [4] **Gandhi, P. J.**, Wetzel, A., Boylan-Kolchin, M., Sanderson, R. E., et al., “*Testing the near-far connection with FIRE simulations: UVLFs, comparison to high-redshift simulations, and distance dependence*” (in prep for submission to ApJ)
- [3] Wetzel, A., Samuel, J., **Gandhi, P. J.**, et al., “*Second public data release of the FIRE-2 cosmological zoom-in simulations of galaxy formation*”, arxiv:2508.06608, 2025
- [2] **Gandhi, P. J.**, Wetzel, A., Boylan-Kolchin, M., Sanderson, R. E., et al., “*Testing the near-far connection with FIRE simulations: inferring the stellar mass function of the proto-Local Group at $z > 6$ using the fossil record of present-day galaxies*”, MNRAS, 533, 1059G, 2024
- [1] **Gandhi, P. J.**, Wetzel, A., Hopkins, P. F., Shappee, B. J., et al., “*Exploring metallicity-dependent rates of Type Ia supernovae and their impact on galaxy formation*”, MNRAS, 516, 1941G, 2022

PUBLICATIONS, CONTRIBUTING AUTHOR

- [12] Mercado, F. J., Baxter, D. C., Rodriguez Wimberly, M. K., et al., (including **Gandhi, P. J.**), “*The Quenched Fraction of Satellites around Simulated Milky Way-Mass Galaxies*”, arXiv: 2512.06071, 2026, (submitted to ApJ)
- [11] Samuel, J., Boylan-Kolchin, M., Feldmann, R., et al., (including **Gandhi, P. J.**), “*Resolving galaxy formation in the early Universe with BonFIRE and CampFIRE*”, arXiv: 2605.24104, 2026, (submitted to ApJ)
- [10] Myrtaaj, O., Bullock, J. S., et al., (including **Gandhi, P. J.**), “*From protogalaxy through thick and thin: Why did the Milky Way evolve in three kinematic phases?*”, arXiv: 2605.22806, 2026 (submitted to MNRAS)
- [9] Moreno, J., Wheeler, C., Mercado, F. J., et al. (including **Gandhi, P. J.**), “*Beyond Cloud-9: The case for discovering more HI-rich failed halos*”, arXiv: 2604.27047, 2026 (submitted to ApJ)
- [8] Kundu, A., Sanderson, R., Lidz, A., **Gandhi, P. J.**, et al., “*Rise of the forsaken relics: connecting present-day stellar streams and phase-mixed galaxies to the Epoch of Reionization*”, ApJ, 999, 124K, 2026
- [7] Meziani, Y. J., Hopkins, P. F., Grudic, M. Y., et al. (including **Gandhi, P. J.**), “*FORGE’d in the Early Universe: The Effect of Protostellar Outflows on Pop III Accretion*”, arXiv: 2602.02953, 2026 (submitted to OJA)
- [6] Moreno, J., Wheeler, C., Mercado, F. J., Rodriguez Wimberly, M. K., et al. (including **Gandhi, P. J.**), “*Igniting galaxy formation in the post-reionization Universe*”, ApJ, 997, 181M, 2026
- [5] Wheeler, C., Moreno, J., Rodriguez Wimberly, M. K., et al. (including **Gandhi, P. J.**), “*How invisible stellar halos bias our understanding of ultra-faint galaxies*”, ApJ, 995, 162W, 2025
- [4] de los Reyes, M. A., Asali, Y., Wechsler, R., (including **Gandhi, P. J.**), “*Stellar Mass Calibrations for Local Low-mass Galaxies*”, ApJ, 989, 91D, 2025
- [3] Horta, D., Cunningham, E. C., Sanderson, R., et al. (including **Gandhi, P. J.**), “*The proto-galaxy of Milky Way-mass haloes in the FIRE simulations*”, MNRAS, 527, 9810H, 2024

- [2] Lemaux, B. C., Fuller, S., Bradač, M., Pentericci, L., Hoag, A., et al. (including **Gandhi, P. J.**), “*The size and pervasiveness of ly-UV spatial offsets in star-forming galaxies at $z \sim 6$* ”, MNRAS, 504, 3662L, 2021
- [1] Pelliccia, D., Strait, V., Lemaux, B. C., Bradač, M., Coe, D., et al. (including **Gandhi, P. J.**), “*RELICS-DP7: Spectroscopic confirmation of a dichromatic primeval galaxy at $z \sim 7$* ”, ApJ, 908L, 30P, 2021

GRANT FUNDING

- Co-I:** (PI Jenna Samuel) – Roman AR (Theory) Grant 19077 (Cycle 1) 2026
The Cosmic Origins of Globular Clusters
- Co-I:** (PI Jenna Samuel) – JWST AR (Theory) Grant 6278 (Cycle 3) 2024
Illuminating the Early Universe with BonFIRE

SUPERCOMPUTING TIME AWARDED

- PI:** NSF Frontera (Texas Advanced Computing Center) – 50k node-hours 2022-23

TELESCOPE OBSERVING AWARDED

- Co-I:** (PI Yasmeeen Asali) Green Bank Telescope (GBT-26A-388) – 38 hours 2026
Do Quenched Satellite Galaxies Retain Gas? Measuring HI in SAGA Satellites

CONFERENCE TALKS & PRESENTATIONS

- Contributed talk*, Near-field cosmology in the era of big data, Toronto, Canada July '26
- Contributed talk*, Spring symposium on the Metal-poor Universe, STScI May '26
- Contributed talk*, FOLIAGE, Wesleyan College Nov '26
- Invited talk*, Gruber Cosmology Prize Symposium, Yale University Oct '25
- Contributed talk*, Santa Cruz Galaxy Workshop, UCSC, CA Aug '25
- Contributed talk*, Galactic Frontiers II, Dartmouth College Jun '25
- Contributed talk*, Tinsley Workshop on early Universe galaxies, Yale U. Oct '24
- Contributed talk*, GalFRESKA conference, UC Riverside, CA Sep '23
- Contributed talk*, Santa Cruz Galaxy Workshop, CA Aug '23
- Poster presentation*, First Light Conference at MIT, MA Jun '23
- Poster presentation*, IAU 377 Symposium in Kuala Lumpur, Malaysia Feb '23
- Contributed talk*, AAS 240 Summer Meeting in Pasadena, CA Jun '22
- Contributed talk*, SAZERAC SIP: Galaxy formation near and far Dec '21
- Contributed talk*, ISM 2021: Structures, Scales, and Star formation May '21
- Senior thesis talk*, Yale University Mellon Forum Feb '18
- Poster presentation*, AAS 231 Winter Meeting in National Harbour, USA Jan '18
- Poster presentation*, Ivy League Undergraduate Research Symposium at UPenn Nov '17
- Contributed talk*, Yale Astronomy Summer Symposium Sep '16

INVITED COLLOQUIA AND SEMINARS

STRAND theory seminar, UC San Diego Astronomy	May '26
Galaxy seminar, UW Seattle Astronomy	May '26
Physics & Astronomy colloquium, Amherst college	Nov '25
Tea talk, Caltech Astronomy tea tal	Apr '25
Lunch seminar, Carnegie Observatories	Apr '25
Galaxy lunch seminar, Yale Astronomy	Apr '24
Tea talk, Stanford KIPAC	Sep '23
Astronomy lunch seminar, Pittsburgh-CMU	Sep '23
Seminar, UT Austin Astronomy	May '23
Group meeting, UPenn Galaxy Dynamics group	Feb '23
Physics & Astronomy colloquium, UC Davis	Oct '22
Carnegie Tea, Carnegie Observatories	Mar '22
Journal club seminar, UH Institute for Astronomy (IfA)	Feb '22

PUBLIC TALKS

<i>Invited</i> , Museum of Science & Curiosity, Sacramento	May '22
<i>Invited</i> , Mt. Diablo Astronomical Society	July '20

RESEARCH (CO-)ADVISING

Graduate Students (2)	
1. Bhavya Pardasani (Yale University)	Jun '25 - Present
2. Yasmeen Asali (Yale University)	Jan '25 - Jun '26
Undergraduate Students (3)	
1. Grace Gogglietino (Colgate University)	Jun '25 - Present
2. Alfredo Calderon (Cal Poly Humboldt)	Jun '23 - Sep '23
3. Heather Pearson (Oberlin College)	Jun '22 - Jun '23

TEACHING

<i>Lead Teaching Assistant UC Davis</i>	
• Astrophysics of Galaxies	Winter '22
• Undergraduate Astronomy Lab	Spring '19
• Introduction to Astronomy	Fall '19
<i>Teaching Assistant UC Davis</i>	
• Physics for Life Sciences	Fall '18, Winter '19, Summer '19, Winter '20, Spring '20
<i>Organiser Sacramento School District Astronomy Olympiad</i>	Jan '19, Jan '20

CONFERENCE ORGANIZING

<i>LOC & SOC</i> , Galactic Frontiers III, Yale University	Mar '27
<i>SOC</i> , FIRE Modelling and Galaxy Evolution (FOLIAGE), Wesleyan U.	Nov '25

SERVICE WORK & OUTREACH

<i>Referee</i> , Open Journal of Astrophysics (OJA)	
<i>Colloquium organizing committee</i> , Yale Astronomy	Jun '26 - May '27
<i>Data Manager</i> , FIRE Collaboration	2023 - 2025
UAW Labor Summer	Jun '23 - Sep '23
<i>DEI Committee co-chair</i> , Astrobites blog (AAS)	Sep '21 - Aug '22
<i>Author</i> , Astrobites blog (AAS)	Dec '20 - Sep '24
<i>Organizer</i> , Astronomy on Tap, Davis (science cafe)	Jan '21 - Sep '24
<i>Chair</i> , UCD Diversity & Inclusion in Physics & Astronomy	Sep '21 - Sep '24
<i>Graduate student rep</i> , UCD Physics/Astronomy Graduate Admissions	Sep '21 - Jan '22
<i>Graduate student rep</i> , UCD Physics/Astronomy Climate Survey	Aug '20 - Apr '21
<i>Member</i> , UCD Diversity & Inclusion in Physics & Astronomy	Sep '19 - Aug' 21

PROGRAMMING

Languages/Platforms: python, C, Java, Unix/Linux/macOS, Windows, slurm

Packages: numpy, scipy, astropy, matplotlib, MPI, OpenMP, GIZMO, Lenstool, SWUnited